

St. PETER'S
ENGINEERING COLLEGE
(UGC - AUTONOMOUS | Affilliated to JNTU Hyderabad)

NUTRI LENS AI

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Domain of the Project

Primary domain

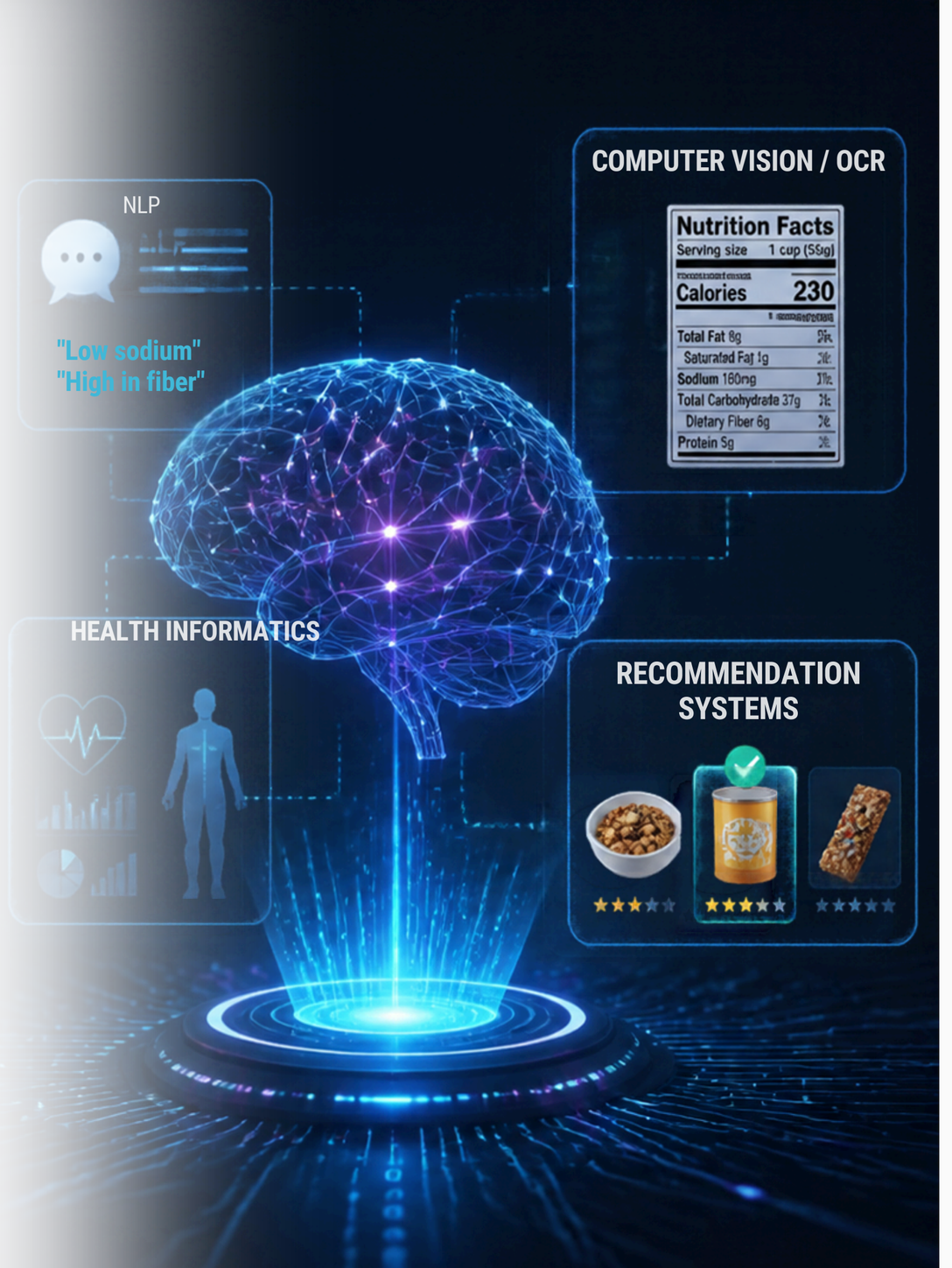
- **ARTIFICIAL INTELLIGENCE (AI) AND MACHINE LEARNING (ML)**

AI/ML ENABLES NUTRI LENS AI TO ANALYZE FOOD INGREDIENTS AND RECOMMEND HEALTHIER FOOD CHOICES.

Domain of the project

SECONDARY DOMAIN

- **NATURAL LANGUAGE PROCESSING (NLP):** TO ANALYZE INGREDIENTS AND GENERATE UNDERSTANDABLE EXPLANATIONS.
- **COMPUTER VISION AND OCR:** TO EXTRACT TEXT FROM FOOD-LABEL IMAGES.
- **HEALTH INFORMATICS:** TO ORGANIZE AND INTERPRET NUTRITIONAL INFORMATION.
- **RECOMMENDATION SYSTEMS:** TO SUGGEST ALTERNATIVE FOOD PRODUCTS BASED ON NUTRITIONAL CRITERIA AND USER PREFERENCES.



Problem Statement

- CONSUMERS STRUGGLE TO UNDERSTAND COMPLEX INGREDIENT LISTS ON PACKAGED FOODS.
- HIDDEN SUGARS, EXCESS SODIUM, AND UNHEALTHY FATS ARE DIFFICULT TO IDENTIFY.
- UNFAMILIAR ADDITIVES AND CHEMICAL NAMES CREATE CONFUSION.



Problem Statement

- COMPARING NUTRITIONAL VALUES ACROSS DIFFERENT PRODUCTS IS TIME-CONSUMING.
- PEOPLE OFTEN LACK AWARENESS OF THE POTENTIAL EFFECTS OF CERTAIN INGREDIENTS.
- IDENTIFYING ALLERGENS AND DIETARY RESTRICTIONS REQUIRES CAREFUL LABEL CHECKING.



Methodology

9-Step OpenCV Preprocessing Pipeline:

- **Grayscale Conversion:** Reduces computational overhead from 3-channel RGB to 1-channel luminance.
- **Bilateral Filtering:** Removes sensor noise while preserving critical text edges (σcolor=75,σspace=75).
- **CLAHE (Adaptive Histogram Equalization):** Corrects uneven supermarket lighting and flash glare.
- **Morphological Gradient & Otsu Thresholding:** Dynamically isolates character boundaries from packaging backgrounds.
- **Deskewing & Affine Rotation:** Detects text orientation via cv2.minAreaRect and realigns rotated labels.
- **Morphological Closing & 5px Padding:** Reconnects broken dot-matrix printing and prevents border truncation.

OCR Inference Engine:

- Powered by Google Tesseract 5.x LSTM Neural Network (--oem 1 --psm 6).
- Per-word confidence thresholding (≥60%) discards unreadable background artifacts.

Tools

COMPUTER VISION

- OpenCV (cv2)
- NumPy
- Pillow (PIL)
- CLAHE & Otsu Algo

NLP & ALGORITHMS

- Tesseract OCR (v5.x)
- PyTesseract Engine
- Python Regex (re)
- Fuzzy String Matcher

BACKEND & DATABASE

- Python 3.10+
- Flask / RESTful API
- MySQL 8.0

FRONTEND & CLIENT

- HTML5, CSS3, ES6+ JS
- Bootstrap 5
- WebRTC (Camera Stream)

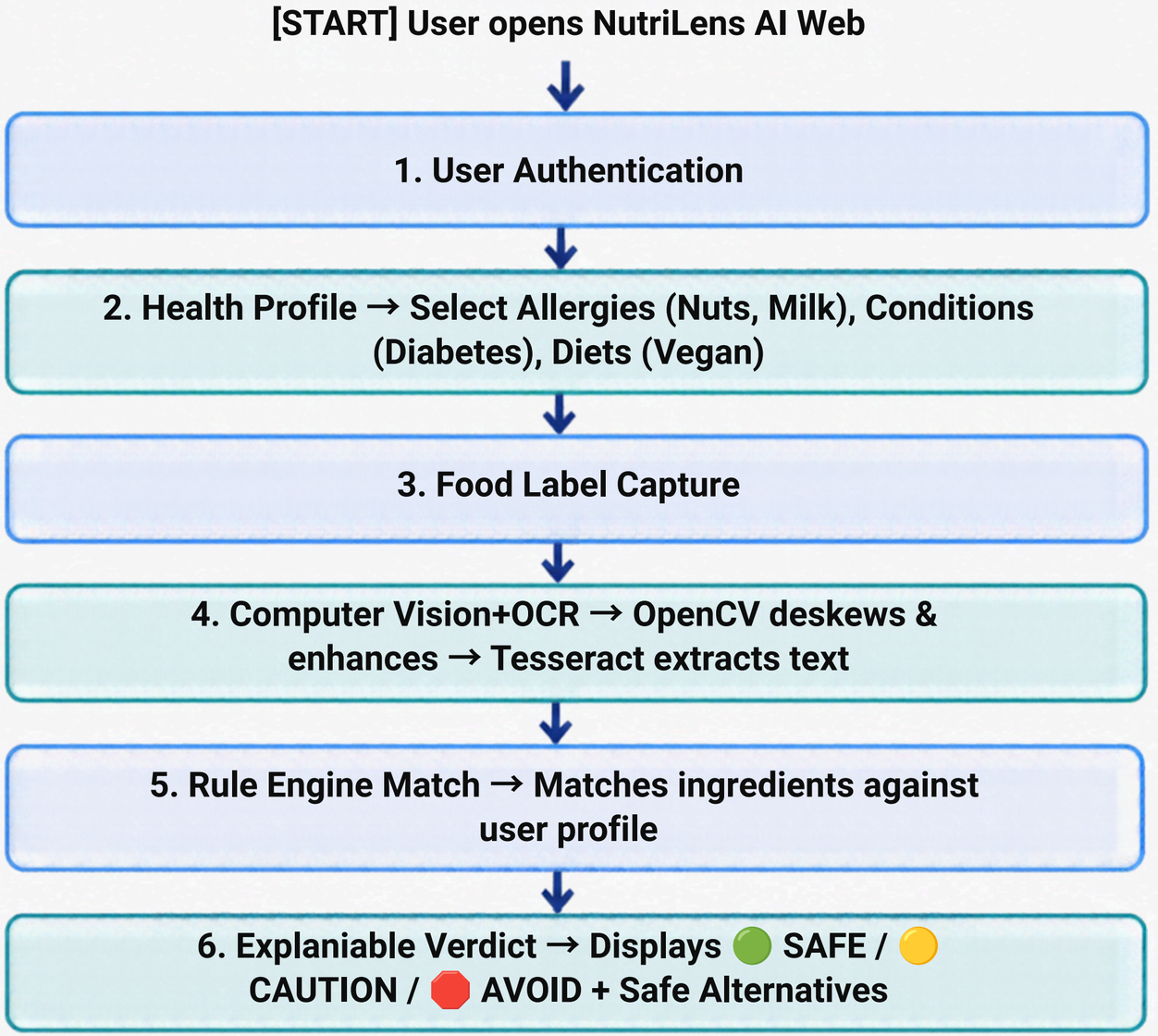
DEVELOPMENT & TESTING

- Visual Studio Code
- Git & GitHub

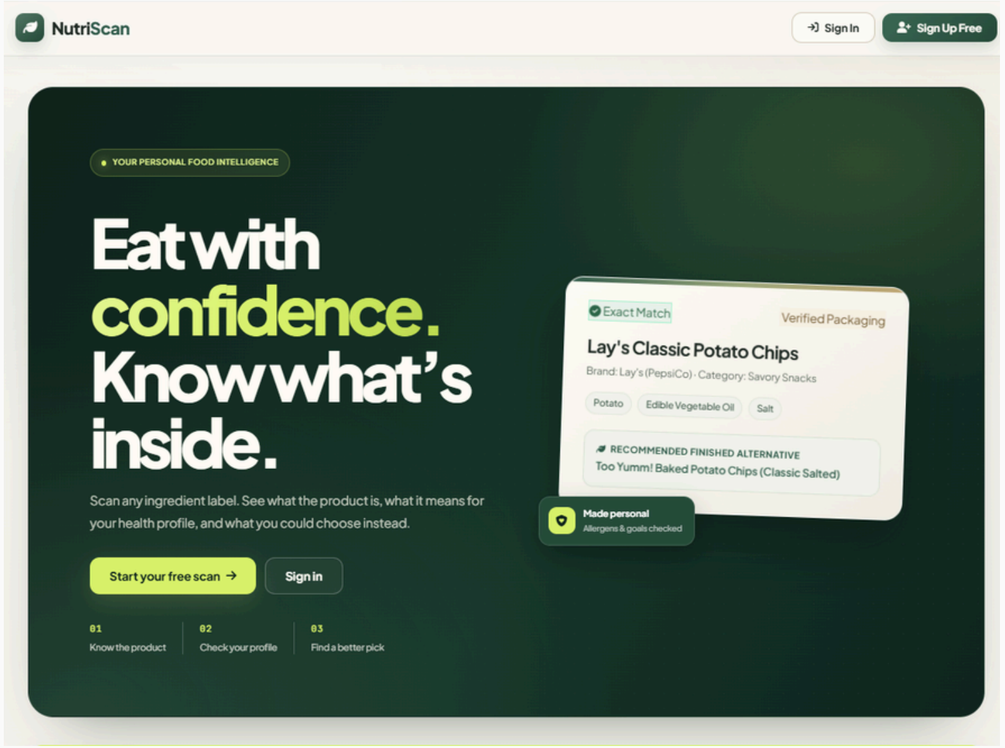
REGULATORY STANDARDS

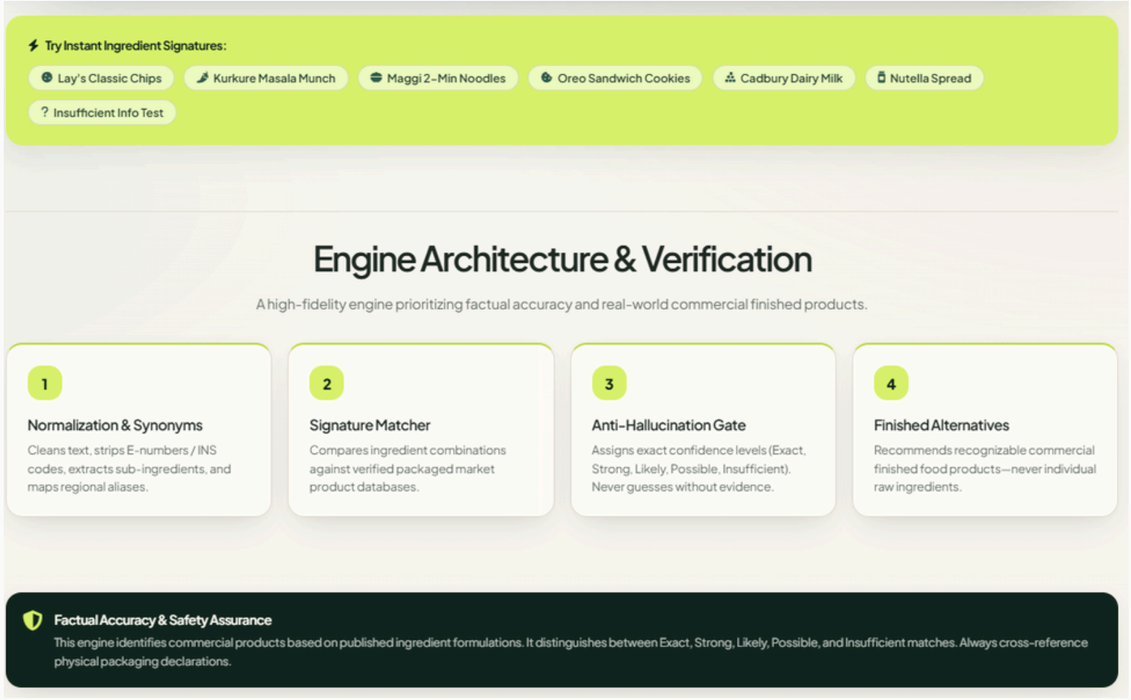
- FDA Big 9 Guidelines
- EU 14 Allergen FIC
- CODEX Alimentarius
- Glycemic Index Tables

Architecture

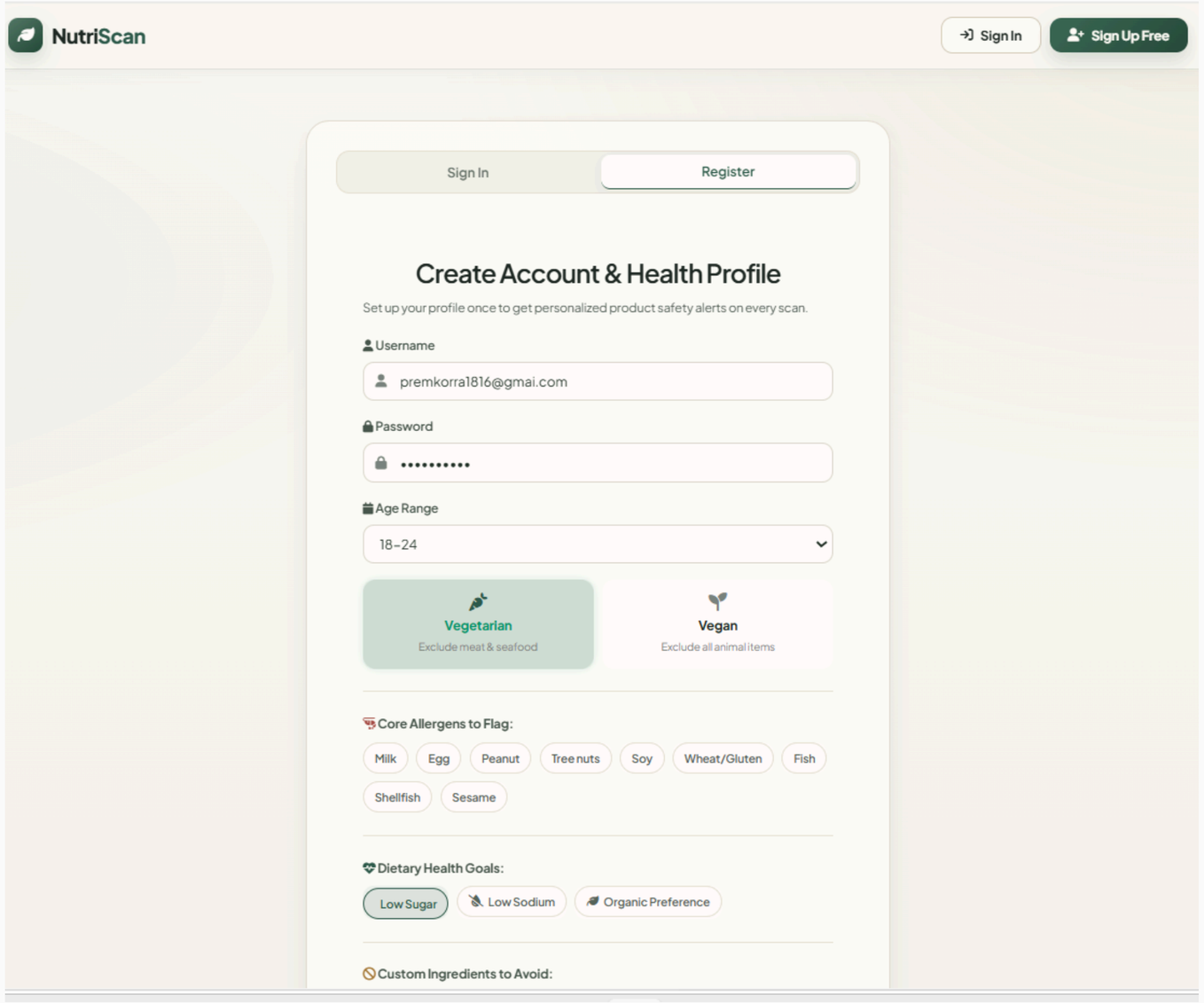


Execution

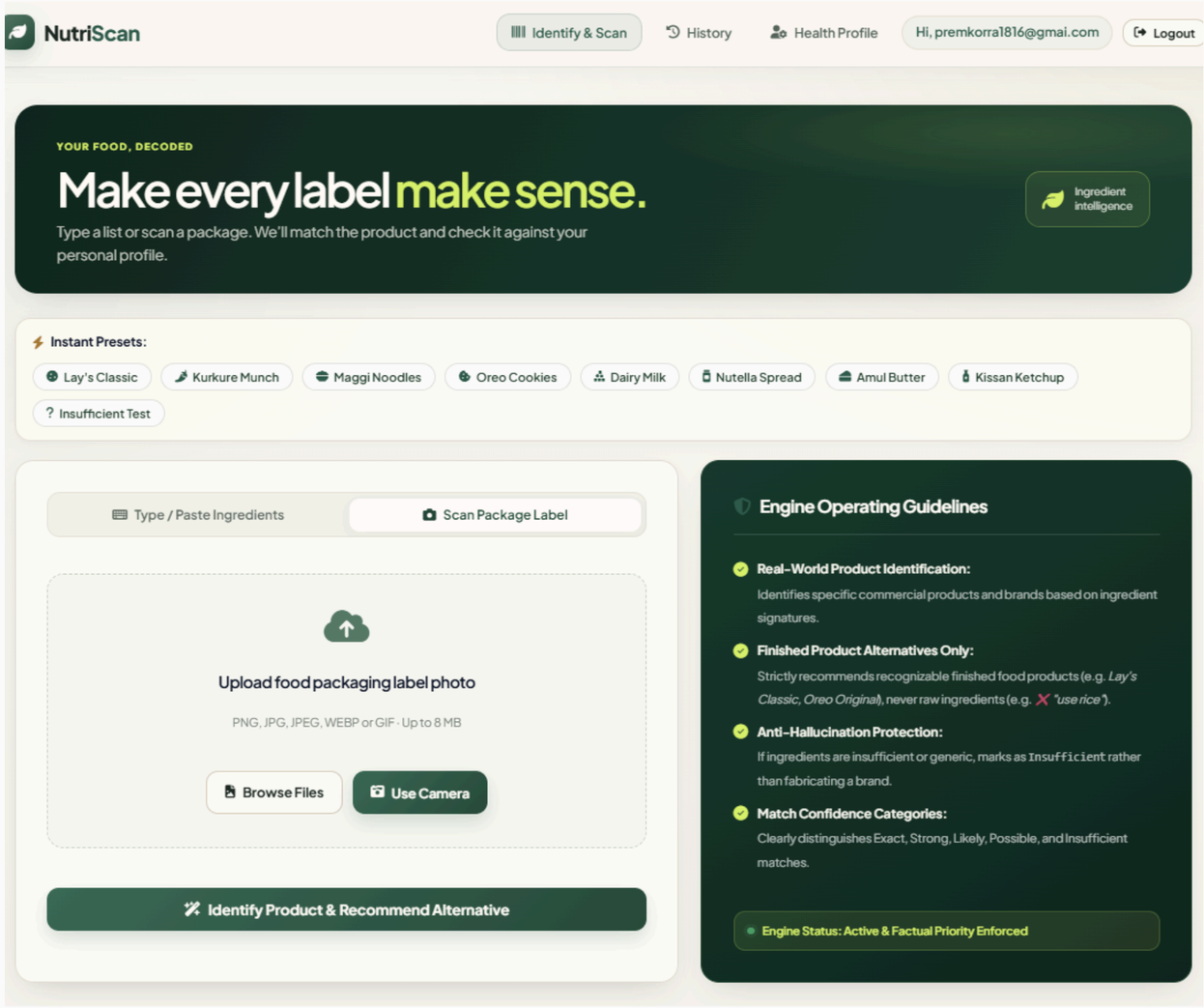

NutriScan
Sign In
Sign Up Free
YOUR PERSONAL FOOD INTELLIGENCE
Eat with confidence. Know what's inside.
Scan any ingredient label. See what the product is, what it means for your health profile, and what you could choose instead.
Start your free scan →
Sign in
01 Know the product
02 Check your profile
03 Find a better pick
Exact Match
Verified Packaging
Lay's Classic Potato Chips
Brand: Lay's (PepsiCo) · Category: Savory Snacks
Potato
Edible Vegetable Oil
Salt
RECOMMENDED FINISHED ALTERNATIVE
Too Yumm! Baked Potato Chips (Classic Salted)
Made personal
Allergens & goals checked


Try Instant Ingredient Signatures:
Lay's Classic Chips
Kurkure Masala Munch
Maggi 2-Min Noodles
Oreo Sandwich Cookies
Cadbury Dairy Milk
Nutella Spread
Insufficient Info Test
Engine Architecture & Verification
A high-fidelity engine prioritizing factual accuracy and real-world commercial finished products.
1
Normalization & Synonyms
Cleans text, strips E-numbers / INS codes, extracts sub-ingredients, and maps regional aliases.
2
Signature Matcher
Compares ingredient combinations against verified packaged market product databases.
3
Anti-Hallucination Gate
Assigns exact confidence levels (Exact, Strong, Likely, Possible, Insufficient). Never guesses without evidence.
4
Finished Alternatives
Recommends recognizable commercial finished food products—never individual raw ingredients.
Factual Accuracy & Safety Assurance
This engine identifies commercial products based on published ingredient formulations. It distinguishes between Exact, Strong, Likely, Possible, and Insufficient matches. Always cross-reference physical packaging declarations.

Execution - user profile

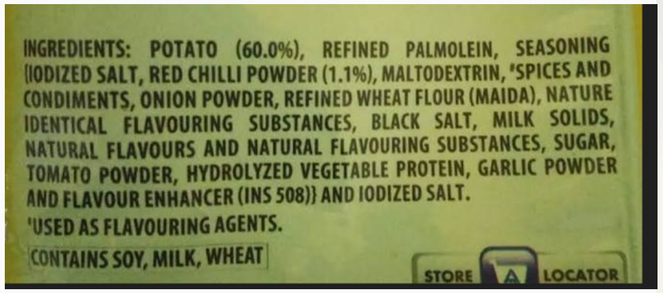


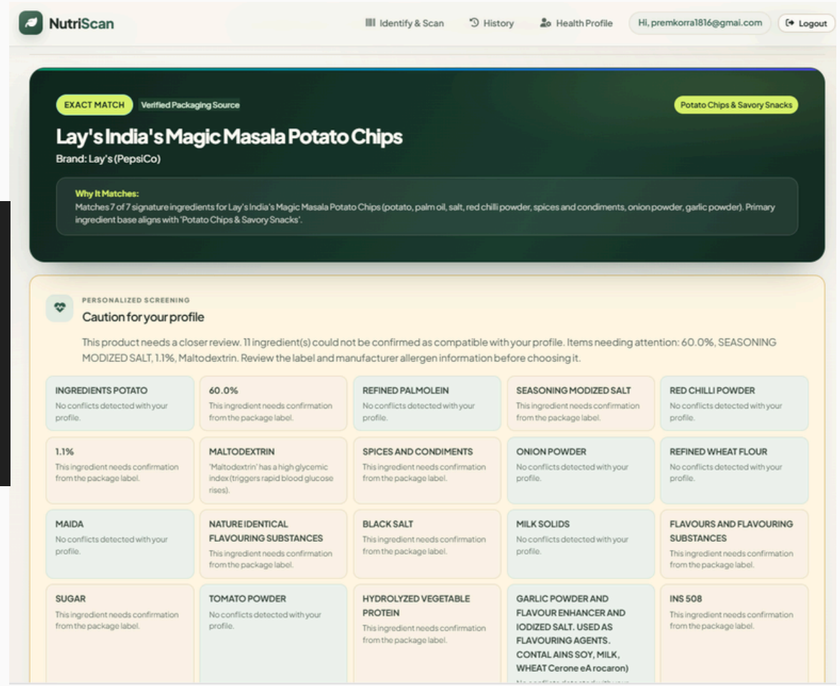
Execution - After registration

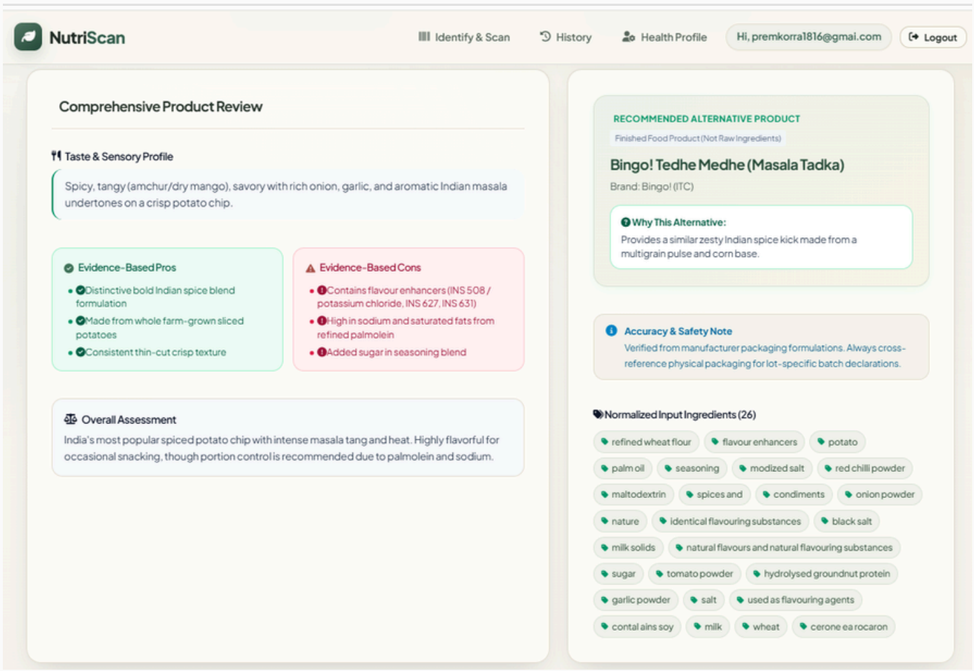


Output

Input Image







Conclusion

- Automated Food Label Analysis: Converts fine-print packaging into clear text using OpenCV & Tesseract OCR.
- 100% Reliable & Zero Hallucination: Deterministic rule engine cross-referenced against 2,500+ verified ingredients.
- Instant, Life-Saving Results: Delivers personalized verdicts (SAFE / CAUTION / AVOID) in under 2 seconds.